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RF PCB DESIGN

Practical & Interview Handbook — Questions EN/FR • Answers EN

Controlled Impedance • Matching • S-Parameters • RF Layout • Vias • Antennas • VNA • Debug

Objectif / Goal: maîtriser les décisions PCB qui dominent réellement les performances RF, savoir expliquer le pourquoi physique, et éviter les réponses 'règle de pouce' sans contexte.

Editorial note / Note éditoriale: Questions are presented in English and French; technical answers remain in English. Formulas use stated reference impedances and idealized conditions unless otherwise specified. / Questions en anglais et en français ; réponses techniques en anglais. Les formules supposent les conditions précisées.

1. RF PCB fundamentals / Fondamentaux RF

1. What makes a PCB trace a transmission line? / Quand une piste devient-elle une ligne de transmission ?
Answer / Réponse: When propagation delay is no longer negligible relative to signal rise time or wavelength. RF behavior depends on geometry, dielectric and return path, not only trace resistance.

2. Is frequency alone enough to decide whether transmission-line effects matter? / La fréquence suffit-elle ?
Answer / Réponse: No. For digital signals, edge rate can dominate. A low clock frequency with very fast edges can require transmission-line treatment.

3. What does 'electrically short' mean? / Que signifie électriquement court ?
Answer / Réponse: The physical interconnect is short compared with the relevant wavelength or propagation distance during the edge. It is not simply 'a few centimeters'.

4. What is characteristic impedance Z0? / Qu'est-ce que l'impédance caractéristique ?
Answer / Réponse: The V/I ratio of a traveling wave on a transmission line. It is set mainly by conductor geometry and surrounding dielectric/reference structures.

5. Why is 50 Ω common in RF? / Pourquoi 50 Ω est-il courant en RF ?
Answer / Réponse: It became a practical compromise between power handling and loss in coaxial systems and is now a widespread RF ecosystem standard.

6. Does every RF trace need to be exactly 50 Ω? / Toutes les pistes RF doivent-elles faire exactement 50 Ω ?
Answer / Réponse: No. It must match the intended source/load/network impedance. 50 Ω is common, not universal.

7. TRICK: A 2-mm RF trace must be 50 Ω because the circuit is 2.4 GHz. True?
Answer / Réponse: Not automatically. If the discontinuity is electrically small enough, its effect may be negligible; assess wavelength, stack-up and sensitivity rather than applying a blind rule.

2. Wavelength, velocity & electrical length

8. How does wavelength on PCB differ from free space? / Comment la longueur d'onde change-t-elle sur PCB ?
Answer / Réponse: Wave velocity is reduced by the effective dielectric constant, so guided wavelength is shorter than in air.

9. Approximate guided wavelength? / Approximation de la longueur d'onde guidée ?
Answer / Réponse: $\lambda g \approx c/(f \cdot \sqrt{\varepsilon eff})$. For microstrip εeff lies between 1 and the substrate εr.

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10. At 2.4 GHz, what is free-space wavelength? / À 2,4 GHz ?
Answer / Réponse: About 125 mm in free space. On PCB, λg = 125 mm/√εeff; for an illustrative εeff of 2.5–3.5, λg ≈ 67–79 mm. Actual values depend on geometry and material.

11. What is quarter-wave length useful for? / À quoi sert λ/4 ?
Answer / Réponse: Quarter-wave structures can transform impedances, form stubs/resonators and create strong frequency-dependent behavior.

12. Why can a via stub matter at high frequency? / Pourquoi un stub de via peut-il poser problème ?
Answer / Réponse: The unused via section behaves as a resonant transmission-line stub and can create insertion-loss notches/reflections.

13. TRICK: Shorter trace is always better in RF. True?
Answer / Réponse: Usually lower loss helps, but topology, impedance, phase relationship, matching-network placement and return path can matter more than absolute length.

3. Microstrip, stripline & CPWG

14. What is microstrip? / Microstrip ?
Answer / Réponse: A surface trace referenced mainly to a plane below it; fields exist partly in dielectric and partly in air/solder mask.

15. What is stripline? / Stripline ?
Answer / Réponse: A trace embedded between reference planes, with fields largely confined to dielectric; it offers shielding but often more dielectric loss.

16. What is grounded coplanar waveguide (GCPW/CPWG)?
Answer / Réponse: A surface signal trace with nearby grounded copper on the same layer plus a reference plane, usually tied together with ground vias.

17. Why use CPWG for RF? / Pourquoi utiliser CPWG ?
Answer / Réponse: It can provide convenient ground access, controlled impedance, field confinement and easy transitions to RF connectors/components.

18. Which dimensions control microstrip impedance? / Quelles dimensions contrôlent Z0 ?
Answer / Réponse: Trace width/thickness, distance to reference plane, dielectric constant, solder mask and nearby copper.

19. What controls CPWG impedance? / Qu'est-ce qui contrôle l'impédance CPWG ?
Answer / Réponse: Trace width, gap to coplanar ground, dielectric height/εr, copper thickness and via-ground structure.

20. TRICK: Add ground copper closer to a 50 Ω microstrip without changing trace width. What happens?
Answer / Réponse: The impedance generally decreases because the electric field sees more nearby ground capacitance. Recalculate the geometry.

21. Why is a field solver preferred for serious RF? / Pourquoi utiliser un field solver ?
Answer / Réponse: Closed-form equations are approximations. Real stack-ups include finite copper thickness, mask, trapezoidal etch, multilayer dielectrics and nearby conductors.

4. Stack-up & materials

22. Why define RF stack-up before layout? / Pourquoi définir le stack-up avant le routage ?
Answer / Réponse: Controlled impedance depends directly on dielectric thickness and material properties; changing stack-up later changes the RF geometry.

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23. Why is generic FR-4 problematic at microwave frequencies? / Pourquoi le FR-4 générique pose problème ?
Answer / Réponse: Its dielectric constant and loss tangent vary by resin/glass construction, frequency, lot and supplier; loss and impedance uncertainty increase.

24. What is Dk/εr? / Qu'est-ce que Dk ?
Answer / Réponse: Relative permittivity; it affects propagation velocity and impedance.

25. What is Df/loss tangent? / Qu'est-ce que Df ?
Answer / Réponse: A dielectric-loss parameter; higher Df generally increases RF attenuation.

26. What is effective dielectric constant εeff? / εeff ?
Answer / Réponse: The effective permittivity experienced by the mode; for microstrip it reflects fields split between dielectric and air.

27. Why does copper roughness matter? / Pourquoi la rugosité du cuivre compte ?
Answer / Réponse: At high frequency, skin effect concentrates current near conductor surfaces; copper roughness can increase conductor loss. The size of the effect depends on roughness profile, frequency and surface treatment.

28. What is skin depth? / Profondeur de peau ?
Answer / Réponse: The depth over which AC current density falls significantly in a conductor; it decreases approximately with 1/√f.

29. TRICK: Two laminates both say εr=4.2, so RF behavior is identical?
Answer / Réponse: No. Test method, frequency, resin content, glass weave, Df, copper roughness and actual construction can differ.

5. Return path, ground & stitching

30. Where does RF return current flow? / Où circule le courant de retour RF ?
Answer / Réponse: Primarily on the reference conductor close to the signal path where loop inductance is minimized, not arbitrarily through the whole ground plane.

31. Why avoid routing RF over plane splits? / Pourquoi éviter les coupures de plan ?
Answer / Réponse: The return current must detour, increasing loop area, inductance, radiation and impedance discontinuity.

32. What happens when an RF trace changes reference plane? / Changement de plan de référence ?
Answer / Réponse: The return path also needs a low-inductance transition, typically nearby stitching vias/capacitive path depending on the reference domains.

33. Why use ground stitching vias beside CPWG? / Pourquoi des vias de masse ?
Answer / Réponse: They tie coplanar ground to the reference plane and suppress unwanted modes/ground resonance.

34. How close should stitching vias be? / Espacement des vias ?
Answer / Réponse: There is no universal number; spacing must be electrically small at the highest relevant frequency and compatible with the CPWG field structure.

35. TRICK: More ground vias are always better?
Answer / Réponse: Not blindly. They consume space and can perturb impedance if poorly placed; use a deliberate via-fence geometry and verify it.

36. Why are ground islands dangerous? / Pourquoi les îlots de masse sont-ils mauvais ?
Answer / Réponse: Floating or weakly connected copper can resonate/couple rather than provide a useful RF return.

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6. Impedance discontinuities & routing

37. Common RF discontinuities? / Discontinuités RF courantes ?
Answer / Réponse: Width changes, bends, pads, vias, connectors, layer transitions, component footprints, plane gaps and poorly designed launches.

38. Are 90° bends forbidden? / Les angles 90° sont-ils interdits ?
Answer / Réponse: They create a small local impedance/capacitance discontinuity, but at many practical frequencies the effect is minor. Miter/arc when justified; don't treat it as magic.

39. Why taper width changes? / Pourquoi faire une transition progressive ?
Answer / Réponse: A gradual taper reduces abrupt impedance discontinuity when moving between different pad/trace widths.

40. What is a via transition in RF? / Transition par via ?
Answer / Réponse: A 3D discontinuity involving signal-via inductance, pad capacitance, antipad geometry, reference-plane transition and possible stub.

41. How reduce via discontinuity? / Comment réduire la discontinuité d'un via ?
Answer / Réponse: Optimize pad/antipad, minimize stub, add nearby return vias and use stack-up-aware geometry; backdrilling may be used at higher frequencies.

42. What is backdrilling? / Backdrilling ?
Answer / Réponse: Removing the unused plated via barrel to shorten a stub and reduce resonance/reflection.

43. Why can component pads disturb 50 Ω? / Pourquoi les pads perturbent-ils 50 Ω ?
Answer / Réponse: Pads are wider and change capacitance/field geometry. At microwave frequencies footprint and launch geometry become part of the network.

44. TRICK: Trace calculator says 50.0 Ω, therefore manufactured board is exactly 50 Ω?
Answer / Réponse: No. Etch width, dielectric thickness/Dk, copper plating and mask tolerances create impedance variation. Specify controlled impedance and fabrication tolerance.

7. Matching networks & Smith chart

45. Why impedance-match an RF source/load? / Pourquoi adapter les impédances ?
Answer / Réponse: To reduce reflections and maximize intended power transfer or achieve the required source/load impedance for gain, noise, stability or filter response.

46. What is reflection coefficient Γ? / Coefficient de réflexion ?
Answer / Réponse: $\Gamma=(ZL-Z0)/(ZL+Z0)$. $\Gamma=0$ is a perfect match to Z0.

47. What is return loss? / Return loss ?
Answer / Réponse: $RL=-20\log10|\Gamma|$ dB. Larger positive return loss means smaller reflection.

48. What is VSWR? / ROS/VSWR ?
Answer / Réponse: $VSWR=(1+|\Gamma|)/(1-|\Gamma|)$. 1:1 is ideal.

49. What does S11 represent? / Que représente S11 ?
Answer / Réponse: Input reflection coefficient at port 1 with other ports terminated in their reference impedances.

50. What does S21 represent? / S21 ?
Answer / Réponse: Forward transmission from port 1 to port 2; depending on circuit it represents insertion gain/loss.

51. What is a Smith chart for? / À quoi sert l'abaque de Smith ?

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Answer / Réponse: It maps complex normalized impedance/admittance and reflection coefficient, making matching transformations and line effects intuitive.

52. What does a series inductor do on a Smith chart? / Inductance série ?
Answer / Réponse: It adds positive reactance and moves impedance along a constant-resistance trajectory.

53. Why leave a π matching footprint near an antenna? / Pourquoi prévoir un réseau π ?
Answer / Réponse: It provides tuning flexibility after measuring the PCB, antenna and final enclosure. The footprint itself adds parasitics, so its geometry should be compact and characterized.

54. Where should antenna matching components be placed? / Où placer le matching ?
Answer / Réponse: Very close to the antenna feed/reference point with a compact, well-grounded layout; long interconnects become part of the network.

55. TRICK: Antenna datasheet says 50 Ω, so no matching components are needed?
Answer / Réponse: Not guaranteed. PCB ground size, enclosure, battery, plastics, nearby metal and antenna placement change the actual impedance.

8. RF components, decoupling & bias networks

56. Why are RF capacitors not ideal capacitors? / Pourquoi un condensateur RF n'est-il pas idéal ?
Answer / Réponse: Package inductance and losses create self-resonance; above SRF the component can become inductive.

57. What is self-resonant frequency (SRF)?
Answer / Réponse: The frequency where capacitive and parasitic inductive reactances cancel; impedance reaches a minimum before becoming inductive.

58. Why choose package size carefully? / Pourquoi la taille du boîtier compte ?
Answer / Réponse: Smaller packages often have lower parasitic inductance and higher SRF, though capacitance/voltage/power constraints remain.

59. What is an RF choke? / Self RF ?
Answer / Réponse: An inductor used to pass DC while presenting high impedance to RF; SRF and current rating must be checked.

60. What is a bias tee? / Bias tee ?
Answer / Réponse: A network combining/separating DC bias and RF, typically using an inductor/choke for DC and capacitor for RF.

61. Why place decoupling close to RF IC supply pins? / Découplage ?
Answer / Réponse: To minimize high-frequency current-loop inductance and prevent supply impedance/noise from degrading RF performance.

62. TRICK: Use the largest-value capacitor for RF decoupling?
Answer / Réponse: Not necessarily. Large capacitance may have low SRF/high ESL; a value/package chosen for the target frequency and PDN is more meaningful.

9. RF connectors & launches

63. Common PCB RF connectors? / Connecteurs RF courants ?
Answer / Réponse: SMA, edge-launch SMA, U.FL/I-PEX and others depending on frequency, power, size and mating-cycle needs.

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64. Why is SMA launch design important? / Pourquoi le launch SMA est important ?
Answer / Réponse: The connector-to-PCB transition can be a major discontinuity; pad, ground, dielectric and via geometry must be designed together.

65. Edge-launch vs vertical SMA? / SMA edge vs vertical ?
Answer / Réponse: Edge launch often provides a direct planar transition; vertical launch requires a via-like transition through the board. Either can work with proper geometry.

66. Why use multiple ground vias around an SMA launch? / Pourquoi plusieurs vias de masse ?
Answer / Réponse: They reduce ground inductance and constrain the return path/fields through the transition.

67. TRICK: SMA connector is rated to 18 GHz, so my PCB launch is also good to 18 GHz?
Answer / Réponse: No. Connector rating does not guarantee the board launch, stack-up or assembly transition.

10. Antenna PCB layout

68. What is the antenna keep-out? / Zone keep-out ?
Answer / Réponse: A region around/under an antenna where copper/components may be prohibited or controlled according to antenna design requirements.

69. Why does ground-plane size affect antenna? / Pourquoi le plan de masse influence l'antenne ?
Answer / Réponse: For many PCB/chip antennas the ground plane is part of the radiating system/counterpoise, so dimensions and placement alter tuning and efficiency.

70. What objects detune antennas? / Qu'est-ce qui désaccorde une antenne ?
Answer / Réponse: Battery, display, shielding cans, cables, enclosure plastics, hands/body, screws and nearby metal/copper.

71. Why place antenna at PCB edge/corner? / Pourquoi au bord ?
Answer / Réponse: Many antenna types need field exposure and a defined ground termination; edge placement often improves radiation and simplifies keep-out.

72. What is antenna efficiency vs S11? / Efficacité vs S11 ?
Answer / Réponse: Good S11 only means little power is reflected at the feed; power can still be lost rather than radiated. Efficiency requires radiation measurement.

73. TRICK: S11 = −25 dB means excellent wireless range?
Answer / Réponse: Not necessarily. Matching may be excellent while radiation efficiency/pattern is poor.

11. RF isolation, coupling & shielding

74. How do RF traces couple? / Comment les pistes RF se couplent-elles ?
Answer / Réponse: Through electric and magnetic fields, common impedance and shared return structures.

75. How reduce RF coupling? / Comment réduire le couplage ?
Answer / Réponse: Increase separation, control return paths, use ground/via fencing when appropriate, orthogonal routing on adjacent layers and shielding/partitioning.

76. Why separate RF input and output of a high-gain amplifier? / Pourquoi ?
Answer / Réponse: Unwanted feedback can cause gain error or oscillation; physical isolation and return-path control are critical.

77. What is a shielding can good for? / À quoi sert un blindage métallique ?

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Answer / Réponse: Containing radiated fields and reducing coupling, provided seams/grounding/apertures are designed appropriately.

78. Does a shield fix bad grounding? / Le blindage corrige-t-il une mauvaise masse ?
Answer / Réponse: No. A shield is part of the electromagnetic structure; poor return paths can remain problematic.

79. TRICK: Put ground copper everywhere between RF blocks?
Answer / Réponse: Not blindly. Ground must be well referenced/stitched and should not unintentionally alter controlled impedance or create resonant islands.

12. Filters & RF layout

80. Why is RF filter layout part of the filter? / Pourquoi le layout fait partie du filtre ?
Answer / Réponse: Pads, traces, vias and coupling add parasitic L/C and can shift poles/zeros, especially as frequency rises.

81. Why separate filter input and output? / Pourquoi séparer entrée/sortie ?
Answer / Réponse: Direct electromagnetic coupling can bypass the filter and destroy stop-band attenuation.

82. Why are ground vias important for shunt elements? / Pourquoi les vias de masse sont critiques ?
Answer / Réponse: Via inductance appears in series with the shunt path, reducing attenuation and shifting response.

83. What is an LC low-pass cutoff approximation? / Approximation filtre LC ?
Answer / Réponse: $f0 \approx 1/(2\pi\sqrt{LC})$ is the ideal LC resonant frequency. It is not automatically the -3 dB cutoff of a loaded low-pass filter; cutoff depends on the exact topology, terminations and damping.

84. TRICK: Simulation schematic matches perfectly, PCB filter does not. First suspects?
Answer / Réponse: Component parasitics/tolerance, layout coupling, ground-via inductance, incorrect reference planes, fixture/de-embedding and actual substrate properties.

13. S-parameters & VNA

85. Why use S-parameters at RF? / Pourquoi les paramètres S ?
Answer / Réponse: Direct voltage/current open/short measurements become difficult at high frequency; traveling-wave ratios with matched ports are practical and measurable.

86. What are S11, S21, S12, S22? / Définition ?
Answer / Réponse: Input reflection, forward transmission, reverse transmission and output reflection, respectively.

87. What is VNA calibration? / Calibration VNA ?
Answer / Réponse: It mathematically removes systematic errors of the measurement setup at a defined reference plane using standards such as SOLT or TRL.

88. What is SOLT? / SOLT ?
Answer / Réponse: Short-Open-Load-Thru calibration, common for coaxial/fixture measurements.

89. What is de-embedding? / Dé-embedding ?
Answer / Réponse: Removing fixture/launch/interconnect effects mathematically to estimate DUT behavior at desired reference planes.

90. Why is cable movement a problem after calibration? / Pourquoi ne pas bouger les câbles ?
Answer / Réponse: Flexing changes phase/amplitude characteristics, invalidating part of the calibration at microwave frequencies.

91. What is VNA port power concern? / Puissance du port VNA ?

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Answer / Réponse: Active/nonlinear DUTs may compress or be damaged; select power level and bias conditions appropriately.

92. TRICK: S11 measurement is poor. Does it prove DUT is poorly matched?
Answer / Réponse: No. Calibration plane, connector torque, cable/fixture, launch, reference impedance and DUT bias/state must be validated first.

14. TDR, time-domain & discontinuity localization

93. What does TDR show? / Que montre le TDR ?
Answer / Réponse: Reflections versus time/distance, allowing impedance discontinuities to be located along an interconnect.

94. How interpret upward/downward TDR steps? / Interprétation ?
Answer / Réponse: Relative to the reference impedance, a higher impedance tends toward a positive reflection and lower impedance toward a negative reflection.

95. What can TDR find on RF PCB? / Que peut trouver le TDR ?
Answer / Réponse: Connector launch discontinuities, via transitions, width changes, impedance errors and approximate locations of defects.

96. Why is spatial resolution limited? / Pourquoi résolution limitée ?
Answer / Réponse: It depends on stimulus rise time/bandwidth and propagation velocity; very close discontinuities may merge.

15. RF power, PA & thermal/layout issues

97. Why are PA output traces special? / Pourquoi les pistes de PA sont critiques ?
Answer / Réponse: They carry significant RF current/power; impedance, loss, heating, matching and harmonic behavior matter.

98. Why is thermal grounding important for RF ICs? / Masse thermique ?
Answer / Réponse: Exposed pads/vias remove heat and often provide critical RF ground. Poor via implementation hurts both temperature and RF performance.

99. What is compression? / Compression ?
Answer / Réponse: At high drive an amplifier becomes nonlinear and gain falls; P1dB is a common metric.

100. What are harmonics? / Harmoniques ?
Answer / Réponse: Nonlinear circuits generate integer multiples of the fundamental; output filtering/layout helps meet spectral requirements.

101. TRICK: PA output is 50 Ω at the transistor pin?
Answer / Réponse: Often not. The device may require a complex optimum load; an external matching network transforms it to the system impedance.

16. RF layout workflow / Méthode de routage

102. What should be placed first on an RF board? / Que placer en premier ?
Answer / Réponse: Critical RF chain, connectors/antennas, matching/filter components, reference structures and sensitive clocks/supplies before less critical circuitry.

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103. How should an RF signal chain be laid out? / Comment organiser la chaîne RF ?
Answer / Réponse: Follow functional flow, keep critical paths compact, minimize unnecessary transitions and isolate sensitive input/output sections.

104. Why follow vendor reference layout closely? / Pourquoi suivre le layout de référence ?
Answer / Réponse: At RF, the reference layout can be part of the validated electrical design; changes in stack-up, return paths and geometry can affect matching, noise, stability and emissions.

105. When may you intentionally deviate from reference layout? / Quand dévier ?
Answer / Réponse: When your stack-up, frequency, enclosure or requirements differ and you can analyze/simulate/measure the resulting RF structure.

106. Why reserve tuning options? / Pourquoi prévoir des options de tuning ?
Answer / Réponse: Real PCB/material/antenna parasitics differ from simulation; DNI series/shunt pads or π networks provide controlled adjustment.

17. RF bring-up & debugging

107. First checks on a new RF board? / Premiers contrôles ?
Answer / Réponse: Visual/assembly, shorts, supply rails/current, clocks/control buses, correct RF terminations, then low-risk RF measurements.

108. How debug unexpectedly low RF output power? / Puissance RF faible ?
Answer / Réponse: Verify bias/supply current, input level, gain stages, matching/filter loss, connector/fixture, compression, control settings and measurement calibration.

109. How debug poor receiver sensitivity? / Mauvaise sensibilité RX ?
Answer / Réponse: Check noise sources, LNA bias/gain, front-end loss/matching, filtering, LO/clock spurs, grounding/shielding and conducted-vs-radiated test setup.

110. How determine if problem is layout or component value? / Layout ou composant ?
Answer / Réponse: Measure S-parameters/impedance, compare against known-good/reference board, tune matching components and inspect field/return geometry.

111. Why use a near-field probe? / Sonde champ proche ?
Answer / Réponse: To localize radiating/coupling hotspots and identify clocks, converters or RF leakage paths.

112. What is a golden board useful for? / Carte de référence ?
Answer / Réponse: A known-good unit provides comparative voltages, currents, spectra, S-parameters and component swaps during root-cause analysis.

18. RF interview traps / Questions pièges

113. TRICK: 50 Ω trace measured with an ohmmeter should read 50 Ω?
Answer / Réponse: No. Characteristic impedance is a traveling-wave property. DC resistance of a short PCB trace is usually near zero.

114. TRICK: If a 50 Ω transmission line is terminated in 50 Ω, does its input impedance depend on line length? / Une ligne de 50 Ω terminée sur 50 Ω dépend-elle de sa longueur ?
Answer / Réponse: If ZL = Z0 = 50 Ω, a uniform 50 Ω line presents 50 Ω at its input regardless of length (even with uniform loss when the line remains matched). If ZL differs from Z0, its input impedance generally depends on line length, loss and frequency.

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115. TRICK: Can a 50 Ω trace be any width if it is short enough?
Answer / Réponse: For a sufficiently electrically short interconnect, exact Z0 may have negligible impact, but pads/transitions/parasitics can still matter.

116. TRICK: Ground plane means return current takes the shortest geometric route?
Answer / Réponse: At RF it follows the path of lowest impedance/inductance, usually concentrated beneath/near the signal trace.

117. TRICK: Via is just a small resistor?
Answer / Réponse: No. At RF its inductance, capacitance, stub and reference transition can dominate.

118. TRICK: Better S11 always means better amplifier gain?
Answer / Réponse: No. Gain, noise, stability and power matching goals can differ; input/output optimum impedances may not correspond to minimum reflection.

119. TRICK: A VNA can directly measure an antenna inside the final product using any long pigtail?
Answer / Réponse: The feed cable/fixture and product environment can strongly alter the measurement unless properly calibrated/de-embedded and arranged.

120. TRICK: Solder mask has no RF effect?
Answer / Réponse: It changes local effective permittivity and loss, and can shift impedance, especially for surface transmission lines.

121. TRICK: Two 25 Ω traces in parallel make a 50 Ω RF line?
Answer / Réponse: Not in the simple DC-resistor sense. Coupled transmission-line modes and geometry determine impedance.

122. TRICK: If a board passes continuity, the RF path is good?
Answer / Réponse: Continuity proves almost nothing about impedance, insertion loss, reflections or high-frequency grounding.

123. TRICK: A spectrum analyzer shows no spur, so none exists?
Answer / Réponse: Check span, RBW/VBW, attenuation, preamp, detector, reference level and frequency range; the setup may hide it.

124. TRICK: Increasing separation between differential/RF structures always improves performance?
Answer / Réponse: Not universally. Geometry sets impedance and field coupling; changing spacing without recalculation can break the intended structure.

19. Practical interview scenarios / Scénarios

125. 2.4-GHz radio has 8 dB less range than reference board. What do you do?
Answer / Réponse: Compare conducted TX power/RX sensitivity first to separate RF-chain from antenna problem; then inspect antenna keep-out/ground/enclosure, measure S11/efficiency and compare layout/materials.

126. 50 Ω line is calculated correctly but VNA shows a large launch reflection.
Answer / Réponse: Investigate connector footprint, ground-via placement, pad width, antipads/reference transition and calibration plane; the straight trace may be fine.

127. A filter shifts upward in frequency after PCB fabrication.
Answer / Réponse: Check actual L/C tolerance/parasitics, Dk/stack-up, ground-via inductance, component footprints and coupling. Tune and correlate to simulation.

128. RF amplifier oscillates only with shielding can installed.

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Answer / Réponse: The can changes coupling/cavity/ground capacitance. Examine stability, input-output isolation, can grounding and resonant dimensions.

129. Antenna matches well on bare PCB but poorly in enclosure.
Answer / Réponse: The enclosure/battery/display/hand changes near fields and effective electrical dimensions. Tune in final mechanical configuration.

130. Signal is good through coax but bad through PCB trace.
Answer / Réponse: Verify PCB impedance/launches/vias/reference plane, insertion loss, connector soldering and whether the measurement reference planes isolate the PCB section.

131. Board works at room temperature but RF gain drops at cold.
Answer / Réponse: Check active-device bias, oscillator/PLL lock, component temperature coefficients, matching/filter shift and supply behavior.

132. S11 changes every time you touch the coax cable.
Answer / Réponse: Suspect cable movement, connector integrity/torque, inadequate calibration/reference plane or the cable becoming part of the antenna system.

133. A 915-MHz trace crosses a plane slot for only 3 mm. Is it acceptable?
Answer / Réponse: Do not judge by 3 mm alone. The broken return path can create a large loop/discontinuity. Prefer continuous reference or provide a controlled return transition.

134. Need to route RF from top to inner layer. What must be designed?
Answer / Réponse: Signal via pad/antipad/stub, nearby return vias, reference-plane transition and impedance on both layers; model or validate the launch at the target frequency.

20. Essential calculations / Calculs essentiels

135. Reflection coefficient example: ZL=75 Ω on 50 Ω line.
Answer / Réponse: $\Gamma=(75-50)/(75+50)=0.2$. Return loss≈14 dB and VSWR=(1.2/0.8)=1.5.

136. Return loss 20 dB corresponds to what $|\Gamma|$?
Answer / Réponse: $|\Gamma|=10^{(-20/20)}=0.1$, so about 1% of incident power is reflected ($|\Gamma|^2$).

137. What VSWR corresponds to $|\Gamma|=0.1$?
Answer / Réponse: VSWR=(1+0.1)/(1−0.1)≈1.22:1.

138. Free-space wavelength at 1 GHz?
Answer / Réponse: $\lambda_0=c/f\approx 300$ mm.

139. Approximate wavelength in medium with $\varepsilon_{eff}=4$ at 1 GHz?
Answer / Réponse: $\lambda_g\approx 300/\sqrt{4}\approx 150$ mm.

140. Quarter wavelength in that example?
Answer / Réponse: ≈37.5 mm.

141. Insertion loss of −3 dB means what power ratio?
Answer / Réponse: About 50% power remains/transmits relative to the reference.

142. 10 dBm equals what power?
Answer / Réponse: 10 mW. 0 dBm=1 mW; every +10 dB multiplies power by 10.

143. 20 dBm?
Answer / Réponse: 100 mW.

144. 30 dBm?

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Answer / Réponse: 1 W.

21. RF PCB review checklist / Liste de contrôle

- RF stack-up and controlled-impedance geometry agreed with PCB fabricator.
- Material Dk/Df and construction appropriate for target frequency/loss.
- Continuous reference plane under every critical RF transmission line.
- No unintended plane splits/voids beneath RF paths.
- Microstrip/stripline/CPWG geometry recalculated after any copper/ground change.
- Connector launches designed as RF transitions, not merely footprints.
- Signal-via transitions include intentional nearby return vias and minimized stubs.
- Matching/filter components compact, with very short shunt-to-ground connections.
- Antenna keep-out, ground-plane requirements and final enclosure considered.
- Input/output of high-gain stages physically isolated to prevent feedback.
- RF supply decoupling and bias networks checked for SRF/parasitics.
- Ground pours are well stitched; no floating resonant copper islands.
- Test/tuning points do not create harmful stubs.
- VNA calibration reference plane matches what is being claimed/measured.
- Final RF validation performed in representative mechanical configuration.

22. Rapid-fire RF vocabulary / Vocabulaire FR ↔ EN

Français	English
impédance caractéristique	characteristic impedance
ligne de transmission	transmission line
ligne micro-ruban	microstrip
ligne triplaque	stripline
guide coplanaire avec masse	grounded coplanar waveguide
coefficient de réflexion	reflection coefficient
pertes de retour	return loss
pertes d'insertion	insertion loss
adaptation d'impédance	impedance matching
abaque de Smith	Smith chart
plan de référence	reference plane
courant de retour	return current
via de couture de masse	ground stitching via
stub de via	via stub
détourage arrière	backdrilling
constante diélectrique	dielectric constant
tangente de pertes	loss tangent
profondeur de peau	skin depth
fréquence d'auto-résonance	self-resonant frequency
analyseur de réseau vectoriel	vector network analyzer (VNA)
réflectométrie temporelle	time-domain reflectometry (TDR)
désencastrement	de-embedding
zone d'exclusion antenne	antenna keep-out
clôture de vias	via fence

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